Ahmad Salehiyan

Ph.D. Candidate | Reinforcement Learning and Optimization

Oklahoma State University | Stillwater, Oklahoma
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Google Scholar | ORCID: 0009-0009-2877-8715 | LinkedIn

Research profile

I develop reinforcement learning and optimization methods for sequential decisions under uncertainty, with applications in maintenance planning, warehouse logistics, and health data.

Available full time from June 2027. Open to relocation anywhere in the U.S.

Education

Ph.D., Industrial Engineering and Management

Oklahoma State University | Fall 2023 to present (expected Summer 2027)

Dissertation proposal: "Sequential Decision-Making Under Uncertainty in Smart Engineering Systems."

Advisor: Dr. Akash Deep

M.S., Applied Statistics

Oklahoma State University | Expected Fall 2026 (concurrent)

Experimental design, statistical programming (R and SAS), data analytics, and statistical inference.

M.S., Industrial Engineering

K. N. Toosi University of Technology, Tehran | 2019 to 2022

Thesis: "Predictive Maintenance of Advanced Industrial Machines Using AI Techniques."

Advisor: Dr. Abdollah Aghaie

B.S., Industrial Engineering

Azad University, Qazvin | 2014 to 2019

Research and industry experience

Graduate Research Assistant | Oklahoma State University

Aug 2023 to present

Research on machine learning, deep reinforcement learning, and stochastic optimization for decision-making under uncertainty. Advisor: Dr. Akash Deep.

Research Assistant | K. N. Toosi University of Technology

Oct 2018 to Apr 2020

Engineered time-domain, frequency-domain, and wavelet-based features from sensor signals for early fault detection in industrial machinery.

Industrial Engineer | Karin Crane Company

Apr 2019 to Oct 2019

Coordinated process improvements and documented operational workflows across production teams. Supported resource planning, scheduling, and quality control.

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Research projects and technical methods

Selected research

Multi-agent fleet maintenance

Dissertation research; manuscript in preparation. I study decentralized maintenance scheduling for heterogeneous fleets with shared resources. The two-phase Safe-MARL approach uses soft constraints during exploration and hard constraints for enforcement. No results are reported yet.

Multi-sensor POMDP maintenance

Lead author of a framework for condition-based maintenance under partial observability. Multiple sensor streams inform belief states and control-limit policies. Belief-state aggregation and state-space compression address high-dimensional observations. TechRxiv preprint (2024); related manuscript submitted to INFORMS Journal on Data Science.

Hierarchical DDQN warehouse picker routing

Lead author. Dynamic programming solves aisle-level routing subproblems and a DDQN agent chooses aisle sequences in two-block warehouses. Reported total travel distance is 3.0% to 15.4% shorter than S-shape and largest-gap heuristics in the evaluated settings. SSRN preprint No. 7052096.

Disease clusters in MIMIC-IV electronic health records

Co-author. Association rule mining and clustering of 179,524 adult patients, with 11,733 mortality cases and 846 distinct diseases. Apriori identified 628 rules with confidence above 50%; t-SNE and k-means grouped the rules. IISE conference proceedings (2025).

Transformer-based intrusion detection

First author of two journal articles. Developed Transformer-based frameworks for intrusion detection in edge and industrial IoT systems and false data injection detection in smart water infrastructure. The intrusion study used WUSTL-IIoT-2021, EdgeIIoTset, and TON_IoT benchmarks.

Technical methods and tools

Reinforcement learning and sequential decisions: DQN/DDQN, MAPPO, MDP/POMDP, Value and policy iteration, Ray RLlib, PettingZoo, Gymnasium.

Machine learning and deep learning: PyTorch, Transformers, GANs, Autoencoders, scikit-learn, Anomaly detection, Clustering, t-SNE, Association rule mining.

Optimization: Mixed-integer and linear programming, Benders and Lagrangian decomposition, Monte Carlo simulation, Pyomo, GAMS.

Statistics: R, SAS, Experimental design, Statistical inference, Hawkes processes.

Data and compute: Python, SQL/PostgreSQL, Apache Spark/PySpark (coursework level), HPC cluster, GPU training, Git, LaTeX, Julia.

Visualization: Power BI, Tableau, Excel.

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Publications

Publication status is stated separately below. Preprints and manuscripts in preparation are not listed as peer-reviewed publications.

Journal articles

A. Salehiyan, N. Serrano, F. Hernando-Gallego, D. Martín, J. V. Álvarez-Bravo. A Novel Evolutionary Optimized Transformer-Deep Reinforcement Learning Framework for False Data Injection Detection in Industry 4.0 Smart Water Infrastructures.

Computers, Materials & Continua, 87(2), 68, 2026.

DOI: 10.32604/cmc.2026.075336

A. Salehiyan, P. S. Moghaddam, M. Kaveh. An Optimized Transformer-GAN-AE for Intrusion Detection in Edge and IIoT Systems: Experimental Insights from WUSTL-IIoT-2021, EdgeIIoTset, and TON_IoT Datasets.

Future Internet, 17(7), 279, 2025.

DOI: 10.3390/fi17070279

Conference proceedings

P. Vaghfi Mohebbi, A. Salehiyan, A. Deep. Disease Cluster Analysis in Electronic Health Records: Insights into Mortality and Comorbidity Patterns.

Proceedings of the IISE Annual Conference & Expo, 1-6, 2025.

Under review

A. Salehiyan, J. Lee, M. Kim, A. Deep. Deviation-Informed Maintenance Planning: Cost Implications of Departures from Sensor-Based Optimal Policy.

Submitted to INFORMS Journal on Data Science, 2025.

Preprints

A. Salehiyan, J. Lee, M. Kim, A. Deep. A Scalable POMDP-based Framework for Real-Time Maintenance Planning using Multiple Sensor Signals.

TechRxiv, 2024.

TechRxiv preprint, 2024.

A. Salehiyan, M. Masae, P. Vichitkunakorn, A. Deep. A Hierarchical Deep Q-Learning Approach for Picker Routing in Two-Block Warehouses.

SSRN.

SSRN preprint No. 7052096.

Read the preprint

In preparation

A. Salehiyan, A. Deep. A Multi-Agent Framework for Scalable Fleet Maintenance Planning under System Constraints.

In preparation (dissertation work).

A. Salehiyan, A. S. Dabas, A. Deep. Deep Reinforcement Learning Framework for Maintenance Planning in Multi-Component Systems with Stochastic Interdependence.

In preparation.

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Teaching, presentations, and service

Teaching experience

ENGR 1412 Introductory Engineering Computer Programming

Graduate Teaching Assistant and Lab Instructor | Spring 2026 and Fall 2026

I provide lab instruction in Excel, prepare and post weekly assignments, grade student work, and lead help sessions to reinforce concepts and guide students through course problems.

Engineering Economics

Graduate Teaching Assistant | Summer 2026

I led course help sessions, explained engineering economics concepts, and guided students through problem-solving.

Conference presentations

Hierarchical Decision-Making for Condition-Based Maintenance of Large-Scale Fleets with Shared Resources

INFORMS Annual Meeting | Nov 1, 2026 | Talk | Session SA20

A Multi-Agent Framework for Scalable Fleet Maintenance Planning under System Constraints

INFORMS Annual Meeting | Atlanta, GA | 2025 | Talk

Disease Cluster Analysis in Electronic Health Records: Insights into Mortality and Comorbidity Patterns

IISE Annual Conference & Expo | Atlanta, GA | 2025 | Talk

A Scalable Algorithm for Condition-Based Maintenance with High-Dimensional Sensor Data

Reliability and Maintainability Symposium (RAMS) | Miramar Beach, FL | 2025 | Talk

POMDP-Based Optimal Maintenance Planning Using Multiple Sensor Signals

INFORMS Annual Meeting | Seattle, WA | Oct 2024 | Talk

POMDP-Based Optimal Maintenance Planning Using Multiple Sensor Signals

OSU Student Research Symposium | Stillwater, OK | 2024 | Poster | 3rd Place Award

Service and leadership

Peer reviewer for academic journals, with seven reviews recorded in ORCID. Computers & Electrical Engineering (6 reviews); Water Research (1 review).

President, Iranian Student Association (IRSA), Oklahoma State University | 2025 to present (second year).

Co-founder and executive, ISA-Alliance, a national network of Iranian student associations.

Award and professional memberships

3rd Place, Graduate Poster Category, OSU Student Research Symposium, 2024.

INFORMS (since 2024); IISE and the IISE Reliability and Maintenance Society (since 2024); SME (Society of Manufacturing Engineers).